

Caesar Cipher

Aim

To implement the Caesar cipher, a monoalphabetic substitution cipher that encrypts text by shifting each alphabetic character a fixed number of positions down the alphabet, and decrypts by shifting back by the same amount. Non-letters are left unchanged and letter case is preserved.

Algorithm

```
1 Function Encrypt(plain_text, shift):
2   cipher_text ← ""
3   foreach character ch in plain_text do
4     if ch is an uppercase letter then
5       offset ← (ch - 'A' + shift) mod 26
6       append ('A' + offset) to cipher_text
7     else if ch is a lowercase letter then
8       offset ← (ch - 'a' + shift) mod 26
9       append ('a' + offset) to cipher_text
10    else
11      append ch to cipher_text                                // leave non-letters as-is
12  return cipher_text
```

```
1 Function Decrypt(cipher_text, shift):
2  return Encrypt(cipher_text, -shift)                        // shift back by the same amount
```

Output

Encrypt — plaintext HELLO, shift 3

Each letter's position (A = 0 to Z = 25) is advanced by 3, wrapping around with mod 26.

Char	Position	final_pos = (pos + shift) mod 26	Cipher char
H	7	$(7 + 3) \bmod 26 = 10$	K
E	4	$(4 + 3) \bmod 26 = 7$	H
L	11	$(11 + 3) \bmod 26 = 14$	O
L	11	$(11 + 3) \bmod 26 = 14$	O
O	14	$(14 + 3) \bmod 26 = 17$	R

Result: HELLO -> KHOOR

Decrypt — ciphertext KHOOR, shift 3

Each letter is moved back by 3 (equivalently, encrypt with shift -3); the + 26 before mod 26 keeps the result non-negative.

Char	Position	final_pos = (pos - shift) mod 26	Plain char
K	10	$(10 - 3) \bmod 26 = 7$	H
H	7	$(7 - 3) \bmod 26 = 4$	E
O	14	$(14 - 3) \bmod 26 = 11$	L

Char	Position	final_pos = (pos - shift) mod 26	Plain char
O	14	$(14 - 3) \bmod 26 = 11$	L
R	17	$(17 - 3) \bmod 26 = 14$	O

Result: KHOOR -> HELLO